

MS4215: Lab 2

Espresso Extraction

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Introduction

The dataset `espresso.csv` contains measurements from 15 espresso shots prepared under varying conditions. The variables are:

Variable	Description
<code>extraction_yield</code>	Percentage of soluble material extracted from the coffee grounds (%)
<code>grind_size</code>	Particle size of ground coffee (m); lower values = finer grind
<code>water_temp</code>	Brewing water temperature (°C)
<code>shot_time</code>	Time for the espresso shot to pour (seconds)

The **outcome variable** is `extraction_yield`. The goal is to understand which brewing parameters drive extraction.

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
#install.packages("car")
library(tidyverse)
library(car)

# In the line of code below, you will need to change the location to where you
# have saved the file
espresso <- read_csv("data/espresso.csv")
```

Questions

- (i) Construct the design matrix

Extract the outcome vector $\mathbf{y}$ and construct the design matrix $\mathbf{X}$: a 15×4 matrix with a leading column of ones (for the intercept) followed by the three predictors. Print $\mathbf{X}$.

- (ii) Compute $(\mathbf{X}'\mathbf{X})^{-1}$

Compute the matrix product $\mathbf{X}'\mathbf{X}$ using `t(X) %*% X`, then invert it using `solve()`. Print the result.

- (iii) Compute $\hat{\beta}$

Use the formula $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ to obtain the estimated coefficients. Write out the fitted model equation.

- (iv) Verify using `lm()`

Fit the same model using `lm()` and `summary()`. Confirm that the estimated coefficients match your manual calculation from (iii). What are the standard errors reported by `lm()`?

- (v) Test model and coefficient significance

Using the `summary()` output from (iv):

- State the result of the overall F-test. Is the model statistically significant?
- For each predictor, state whether it is significant at the 5% level.

- (vi) Check multicollinearity

Compute variance inflation factors using `vif()` from the `car` package. Are any VIFs a concern? Does the result change your interpretation of the non-significant predictor found in (v)?